

RESPONSE TO THE REFEREES: NEURAL BELLMAN OPERATORS

QIAN QI
Peking University

Revision R24 responds to both reports on the sealed R23 manuscript, with new proofs and a prospective 110-trajectory study. Forty scientific and technical findings are mapped to the delivered changes. The original full-state objective and all historical material remain. Numerical requirements not established by the delivered evidence are explicitly distinguished from resolved definitional and analytical questions.

KEYWORDS: Stochastic control, Neural Bellman operators, Preference adjustment, Validated numerics, Policy verification, Computational economics.

1. REVISION AND REVIEW LINEAGE

We thank the referees for distinguishing a successful financed policy from a complete numerical-method result. R24 retains the title, economic model, original full-state .01 objective, previous derivations and adverse outcomes. It adds an explicit augmented-state Bellman formulation, momentum-correct output geometry, a conditional financing-gradient bound, disjoint tuning, direct preconditioners, quadrature refinement, certified checkpoints, a four-cell cover and an analytic reference.

The reviewed manuscript is commit `c09835c92042cb881d0bbdd3cab99b98d0a9d225`. We answer the first report at `faffff2ccfa218b5de4ea73a4d0eb21d468166c98` and the second pass at `fdc60e04d6aca33ad964da77db69c8ba3621a14b`. The new branch descends from the second pass. Protocol `b54afa8c9107d70094a1d84be4a3c0279916530c` predates the new scientific execution. There are 110 trajectories and 286 prescribed checkpoints, not another selected replay of the four old endpoints. The following sections map all forty scientific/technical finding identifiers; the accompanying disposition file makes no blanket claim of closure.

2. RESPONSES TO THE FINDINGS

2.1. *State and Bellman identity*

R23-SP-F1, R23F-F2, R23F-F12. The main paper now defines the controller as Markov in (t, u, x, Y, λ) , with frozen expert coefficients. Initial weights and the common pricing factor are retained information, not functions asserted to depend on (t, u, x) alone. Expert wealth is a conditional-price function $F_i(t, Y)$, so independently simulated stochastic subaccounts are unnecessary. We prove the invariant aggregate equations, feasibility, the augmented generator including all mixed covariance terms, and the Bellman semigroup with the original stopping payoff. Under the original completed Brownian filtration, causal deterministic memory does not enlarge the admissible information class, giving equality of unrestricted augmented and original values. This is a formal reformulation with the same economic optimum, not evidence that a particular regional expert policy has full-domain accuracy or can be compressed into the original state.

2.2. The neural-induced geometry

R23-SP-F2, R23F-F5, R23-SP-T2, R23T-T4. Finite-node class equality is retained; neural representational superiority is not inferred. The new transport theorem expands Adam’s actual bias-corrected momentum into historical terms $J_n D_n^{-1} J'_r g_r$ and bounds the nonlinear output remainder. The current matrix $J_n D_n^{-1} J'_n$ is not substituted for that history without additional conditions. Actual trained Jacobian spectra and adaptive-metric spectra are recorded, separately from the old exact interpolation determinant. Direct diagonal, tangent and whitening charts use the same financed quotient, with explicit eigenvalue regularization, normalization and preprocessing costs. A parameter-chart proposition explains invariant first-order output directions under a right inverse, but the corresponding rank condition is not presumed for ill-conditioned trained networks. The equivalent layer scaling leaves initial outputs unchanged rather than changing tanh arguments.

2.3. Tuning rather than nominal learning-rate matching

R23-SP-F3, R23F-F3, R23T-T1. Each first-order arm receives six tuning trials: two seeds disjoint from evaluation, rates .005, .02, .08, and 200 objective/gradient calls per trial. Selection uses mean independently certified lower payoff, with explicit tie and infeasibility rules. Evaluation never retunes. All arms select .08, the grid’s upper endpoint; the manuscript states that this is a finite declared calibration family, not optimal calibration over unspecified rates. All trial costs and outcomes remain in the supplement, including the cost of rates not selected. Equal tuning calls are not labeled equal work.

2.4. What the quotient removes

R23-SP-F4. The old direct objective already centered the intercepts. Dropping the common-intercept coordinate therefore isolates a financing gauge, not a general conditioning advantage. New anisotropic direct charts test additional geometry without changing the attainable family. In the deterministic reference there is no financing gauge: all 48 coordinates, including the consumption intercept, are retained. The unused slope/drift directions are explicitly inactive, not silently discarded. Regression tests check these distinctions.

2.5. The accuracy–work comparison

R23-SP-F5, R23F-F4. The new tables retain all five calibrated first-order methods and both L-BFGS-B representations, with calls, generation time, all checkpoint checks, preprocessing, instrumentation, tuning and upper-library allocation separated. At the coarse rule, neural Adam’s worst deployed bound is below .001925713, while direct L-BFGS-B is below .001923443, with mean generation .321 versus 1.303 seconds. This direct advantage is prominent, not omitted. Neural L-BFGS-B and whitening failures also remain. Dual-free payoff intervals isolate matched-policy ordering from dual slack. Neither a local first-order advantage nor an equal-call endpoint establishes universal neural frontier dominance.

2.6. The original full-state objective

R23-SP-F6, R23F-F1. The original uniform .01 objective is unchanged. The canonical versioned record distinguishes the discounted all-start-time bound 7.241462443133396 from the older undiscounted 7.386001245971047. We preserve the full-state actor and witness rather than replacing them by an augmented expert decoder. R24 does not establish a new full-domain

actor certificate or separate actor payoff improvement, and does not describe regional bounds as orders-of-magnitude full-state progress. The residual-correction construction and earlier derivations remain the constructive foundation for this stronger numerical requirement. State augmentation resolves the information-state ambiguity, not that accuracy requirement.

2.7. *Proposal quadrature and the certified objective*

R23-SP-F7. Evaluation is crossed with orders (8, 16) and (12, 24), while the final stopped-model checker is unchanged. Both rules evaluate gradients at each identical frozen raw policy; root offsets and conditioning diagnostics remain. The reduced-gradient theorem accounts for root error, inverse derivative conditioning, price derivatives and payoff derivatives. It requires uniform derivative hypotheses not established by a two-rule difference. The largest absolute raw-gradient difference is approximately 4.410×10^{-7} , but the largest relative difference is about 22.993 near a fine-rule stationary point; we therefore do not infer continuous-gradient accuracy from a small absolute difference. Preference clipping defines an integrable proposal tail extension without crossing CRRA's singularity at one. No held-out checkpoint node is clipped, and the stopped economy is not changed.

2.8. *The operational acceptance rule*

R23-SP-F8, R23F-F11. Snapshots are prospectively fixed, then certified serially. Deploy only when the candidate lower payoff exceeds the incumbent upper payoff. Among 140 held-out checkpoint decisions, 121 replace the incumbent and 19 reject: seven repeat the preceding candidate and twelve concern changed candidates. The proposal optimizer continues independently of the replayed gate; the study is not rollback-coupled policy iteration and the gate is not credited with driving proposal convergence. The historical 72/72 acceptance finding and distinct rollback stress test are retained with their different roles. Curves reconstruct the cost of the checking schedule, not measured online deployment time.

2.9. *What is held out*

R23-SP-F9, R23F-F6, R23F-F10. R23 is identified as held-out-seed replication after inspecting R22, not independent validation of the method family. R24 was motivated by those known findings but fixed its new design before execution and did not change its selection rule on evaluation results. Explicit preconditioners, quadrature refinement, a larger multicell region and an analytic reference add dimensions beyond seed repetition. Two evaluation seeds are not claimed to establish broad stochastic robustness across architectures or economies. No adverse prescribed configuration is dropped.

2.10. *Certified anytime records*

R23-SP-F10, R23-SP-T7. The fixed caps are 25, 50, 100, 200, 400. A line-search checkpoint uses the latest accepted iterate available by the cap, not an unaccepted or future iterate. Early termination is carried forward with actual calls. Every held-out checkpoint appears in the printed ledger and all 286 checkpoints in machine-readable records, with objective histories, gradients, raw outputs, certificates and decisions. Curves show calls, generation time and reconstructed cumulative certification cost. There is no outcome-selected checkpoint omission.

2.11. *State coverage beyond four vertices*

R23-SP-F11, R23F-F7. A 3×3 grid over $[1.98, 2.02] \times [1.24, 1.27]$ extends the wealth span by 50 percent and generates nine distinct experts per method. All four cells are certified via the retained common-shock concavity and stopping correction, with the explicit unrestricted upper evaluated at new vertices. The neural union bound is .004280526; direct L-BFGS-B gives .004274863. These are continuum regional bounds for initialized augmented policies, not nine sampled-point assertions. Costs use actual expert generation and checks, not duplicated-expert timing. This is not high-dimensional scaling evidence. Improved vertex payoffs alone are not used to infer uniform mixture payoff improvement.

2.12. *Solver error; checker width and dual cost*

R23-SP-F12, R23F-F9, R23T-T7. Results distinguish policy-to-policy intervals, policy enclosure width, the unrestricted upper and every cost component. The 415.789-second historical dual cost is displayed in unamortized and reusable accounting, with its different-machine qualification. In the original economy, upper-minus-lower gap equals dual slack plus actual regret plus lower-enclosure slack. Tight payoff enclosures alone do not identify the first two terms. The analytic reference isolates them where the optimum is known. The preceding near-constant direct regret level is not simply declared a verification floor.

2.13. *Exact-price brackets and delivered-root accuracy*

R23-SP-T1. The continuous exact-price bracket remains an existence and conditioning certificate, not proof of proximity of a delivered finite-rule offset. Delivered feasibility uses independent continuous price and reserve enclosures. The gradient theorem states the additional price and derivative errors required for reduced-gradient control. Analytical hypotheses, frozen-object directed instantiations and ordinary diagnostics are listed separately in the dependency map.

2.14. *Calls are not work*

R23-SP-T3, R23T-T3. Current text uses equal objective/gradient-call cap rather than equal work. Per-call and preprocessing costs differ across representations. Tables and curves retain measured generation, verification, initial checks, instrumentation and tuning. Shared computations are charged conservatively in standalone columns, whose sum is not represented as observed shared-study wall time. Historical prose remains unchanged under an explicit historical interpretation notice.

2.15. *The seed-23203 L-BFGS-B failure*

R23-SP-T4, R23T-T5. The original failure remains. Regeneration reproduces the .010503068 bound. An equivalent powers-of-two layer chart lowers it to .002554827 within its declared 400 calls. Tighter tolerance and direct restart from the saturated failed outputs leave the bound near .010503067. The ledger includes gradients, Jacobian spectra, output distances, calls and termination messages. This distinguishes network-coordinate effects, decoder saturation and trajectory dependence without claiming separately identified causes. The diagnostics are post-hoc and excluded from tuning and held-out method selection; restart budgets are additional.

2.16. *Transitive scientific reproduction*

R23-SP-T5. The package contains readable source, runtime-imported repository-module hashes, inherited-data hashes, numerical versions, protocol identity and generation commands. Rebuilding tables from frozen records is distinguished from fresh regeneration in a disposable checkout. The primary run completed tuning, held-out and multicell stages, then encountered strict JSON encoding of an unused infinite root diagnostic in the no-root reference economy. A metadata-only null encoding fixed this without changing scientific parameters. The original archive remains, and all primary result bytes were checked unchanged before completing reference/failure phases. Publication auditing additionally preserves every reviewed Git blob and locks transitive TeX inputs and PDFs.

2.17. *A reference with a known optimum*

R23-SP-T6. The reference optimum is $c_* = (\sqrt{6} - 1)/2$, reserve c_*^2 , value $-1/c_* + 2 \log c_*$. Strict concavity and a scalar first-order condition prove it. Both classes contain the constant optimum; exact slab integration removes temporal quadrature error. Directed intervals identify the optimization gap. Direct L-BFGS-B reaches a worst gap below 3.393×10^{-13} with width around 10^{-14} . A neural Adam seed remains near .002772, which is retained. This separate economy is not counted as a full-state or nonreplicable stochastic solution.

2.18. *A complete method definition*

R23-SP-T8. One main-text algorithm table specifies information, generator, proposal quadrature, tuning, financing, deployment acceptance, stopped-payoff verification, unrestricted comparator and cost/reuse assumptions. The exact Bellman semigroup, proposal optimizer and acceptance map are distinct objects. The state/value theorem and policy-specific comparison relate them without conflating them under the word operator.

2.19. *Versioning the full-state summary*

R23T-T2. The new schema-version-two record names the discounted bound 7.241462443133396, identifies the misleading historical field and its 7.386001245971047 undiscounted value, and records that neither a new actor payoff improvement nor the .01 full-state target is established. Old JSON files remain byte-identical; the correction does not rewrite historical evidence.

2.20. *Hypotheses and levels of evidence*

R23T-T6, R23T-T8. The supplement lists financing, common-shock coupling, concavity, stopping-range, convex-upper and directed-arithmetic requirements. Theorems, directed checks for frozen policies and ordinary spectra/refinement diagnostics are distinct. MPFR replay remains an arithmetic cross-check of the same model mathematics, not an independent economic derivation. The reference provides a separate mathematical calibration for its own economy.

2.21. *Execution without exact replication*

R23F-F8. The general residual-correction cone and original current-state framework remain and do not require this exact-replication decoder. The new multicell experiment still uses the favorable decoder; the deterministic reference isolates error components rather than testing an unspanned feasibility mechanism. We do not claim a successful nonreplicable stochastic execution in R24. That stronger numerical requirement remains explicit with the unchanged original objective, rather than being relabeled as solved by a different model.

3. PUBLICATION REQUIREMENTS AND EDITORIAL POINTS

The reports' state-definition requirements are answered by the explicit augmented policy, Bellman problem and unrestricted-value identity. They do not establish original-state numerical accuracy. Tuned-frontier requests receive disjoint calibration, direct preconditioners and all checkpoint curves; performance disadvantages remain visible. Quadrature, checkpoint, reference and multicell requests receive new executed evidence within its stated scope. The request for successful nonreplicable stochastic execution is not marked completed.

The abstract states stored weights and factor memory. Current text qualifies seed replication, nominal settings versus calibration, calls versus work, gauge removal versus conditioning, and non-neural upper construction. All original R23 components and adverse outcomes are retained in historical appendices/supplements. The new original-state full-domain certificate, separate full-state payoff improvement, rigorous numerical stopped-gradient error and broad high-dimensional superiority are not asserted. This is a substantive revision with additional proofs and experiments, not a declaration that all publication-level requirements are closed.

4. KEY NEW NUMERICAL TABLES

TABLE I
HELD-OUT CENTRAL STATE: ACCURACY AND COMPUTATION

Rule	Arm	Regret upper	Calls	Gen. s	Check s	Cert. cost s
8/16	Neural-Adam	0.001925713	400.0	1.303	10.379	13.744
8/16	Direct-Adam	0.002039228	400.0	1.181	10.413	13.656
8/16	Diagonal-Adam	0.002008735	400.0	1.179	10.330	13.574
8/16	Tangent-Adam	0.002157232	400.0	1.175	10.333	13.573
8/16	Whitened-Adam	0.01874203	400.0	1.175	10.377	13.617
8/16	Neural-L-BFGS-B	0.009608327	251.0	0.792	10.478	13.333
8/16	Direct-L-BFGS-B	0.001923443	111.5	0.321	10.400	12.784
12/24	Neural-Adam	0.001925705	400.0	1.997	10.370	14.427
12/24	Direct-Adam	0.002039189	400.0	1.859	10.352	14.272
12/24	Diagonal-Adam	0.002008701	400.0	1.853	10.338	14.253
12/24	Tangent-Adam	0.002157182	400.0	1.847	10.377	14.286
12/24	Whitened-Adam	0.01874227	400.0	1.878	10.480	14.420
12/24	Neural-L-BFGS-B	0.009608181	243.5	1.183	10.479	13.722
12/24	Direct-L-BFGS-B	0.001923667	112.0	0.520	10.457	13.037

Regret is the worse of two seeds at $(t,u,x)=(0,2,1.25)$; calls and times are means. Policies are final deployed checkpoints. Checking includes all five scheduled checks, including converged repeats. Certified cost adds preprocessing and initial checking. Diagnostic-only calls and tuning are separate.

TABLE II
FOUR CELLS AND NINE DISTINCT EXPERTS PER METHOD

Arm	Region regret upper	Calls	Gen. s	Check s
Neural–Adam	0.004280526	3600	11.755	18.686
Direct–Adam	0.004398790	3600	10.553	18.638
Tangent–Adam	0.004508256	3600	10.548	18.659
Direct–L-BFGS-B	0.004274863	1250	3.592	18.694

Region $[1.98, 2.02] \times [1.24, 1.27]$ at time zero. Every cell is certified using its deployed expert records and the retained concavity/exit theorem. The controller has the explicit augmented state. Generation/checking are totals across nine genuinely distinct experts; initial checks and preprocessing remain in the ledger.

TABLE III
ANALYTIC REFERENCE: SOLVER GAP AND ARITHMETIC WIDTH

Arm	Gap upper	Payoff width	Calls	Gen. s	Check s
Neural–Adam	0.002771190	9.769963×10^{-15}	400.0	0.321	0.066
Direct–Adam	0.00005492341	1.021406×10^{-14}	400.0	0.198	0.066
Diagonal–Adam	0.00006450277	1.021406×10^{-14}	400.0	0.197	0.066
Tangent–Adam	0.00001955686	1.021406×10^{-14}	400.0	0.197	0.067
Whitened–Adam	0.5646302	1.332268×10^{-14}	400.0	0.196	0.069
Direct–L-BFGS-B	3.392842×10^{-13}	1.021406×10^{-14}	14.0	0.007	0.067

Worst gap and width across two seeds, mean calls and times. This separate deterministic economy has zero class and temporal quadrature errors. Rates transfer from original-economy tuning without reference-specific calibration. The common consumption intercept is retained; the other two output channels are inactive.

TABLE IV
POST-HOC SEED-23203 FAILURE DIAGNOSIS

Case	Regret upper	Calls	Raw distance	Financed distance
regenerated	0.01050307	92	299.963	2.691
scaled	0.002554827	400	186.416	7.409
restart tighter	0.01050307	43	296.812	2.753
direct restart	0.01050307	4	299.963	2.691

Distances are to the unchanged frozen R23 direct solution. Tighter and direct restarts use additional budgets. Regenerated and chart-scaled trajectories start at identical outputs; these diagnostics are excluded from tuning and held-out selection.